

RUTHVIK VISHWANATHWAR

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SUMMARY

Senior Data Engineer and **Databricks Certified Data Engineer** with 11 years of experience designing enterprise-scale lakehouse architectures that enable analytics, compliance, and AI-driven decision-making across healthcare and financial services. Expertise in multi-source data orchestration, governance frameworks, and establishing data quality as organizational discipline. Proven track record architecting systems that reduce time-to-insight by 60%+ and position heterogeneous enterprises for ML-driven operations while maintaining HIPAA/PHI controls and audit accountability.

CERTIFICATIONS

Databricks: Databricks Certified Data Engineer Associate, Platform Architect, Certified Platform Administrator, Generative AI Fundamentals, AI Agent Fundamentals, Lakehouse Fundamentals | dbt fundamentals | SQL for Data Science

PROFESSIONAL EXPERIENCE

Senior Data Engineer | *Frisco, TX (Client/s; Sentara Health, Georgia Pacific)*

May 2022 – Present

- Migrated 40+ legacy SSIS and SQL Server ETL workflows into modular Databricks PySpark and Spark SQL jobs on a Medallion Delta Lake setup, which brought pipeline runtimes down by about 60%.
- Built a reconciliation framework that handled field mappings, crosswalk logic, business rules, and configurable tolerance thresholds. Tuning those thresholds with real data cut our false positives by more than 60%.
- Turned the reconciliation database into a reusable data asset for ML work by capturing quality checks, past exception patterns, and lineage in Delta Lake, so we could start catching data issues before they hit downstream.
- Handled performance and cost tuning across large tables using Z-ordering, partitioning, OPTIMIZE/MERGE, and cluster policies. Query times dropped up to 70%, and one batch job went from 14 hours to under 3.
- Set up governance and CI/CD standards for our HIPAA/PHI healthcare pipelines using masking, RBAC, tokenization, Unity Catalog, and Structured Streaming, and ran design reviews and mentored the other engineers along the way.

Stack: Azure Databricks, PySpark, Spark SQL, ADF, Delta Lake, Delta Live Tables, SQL Server 2019, Azure Synapse, Power BI, SSIS, Python

Senior Data Engineer | *Frisco, TX (Client/s; Blue Cross Blue Shield of Kansas)*

May 2021 – May 2022

- Implemented data validation and reconciliation checks across source-to-report pipelines, improving trust in eligibility, claims, and provider metrics by identifying missing records, duplicate claims, and inconsistent member attributes before executive reporting.
- Designed orchestration patterns across ADF, Databricks Jobs, and Airflow DAGs, standardizing retries, dependency handling, SLA monitoring, and backfill operations.
- Optimized Databricks workload efficiency through partition strategy, file compaction, job tuning, and cluster configuration standards, reducing compute waste and improving pipeline scalability across high-volume datasets.
- Tuned high-volume SQL workloads through indexing strategy, execution-plan analysis, and stored procedure refactoring, reducing report refresh time by 40% and improving dashboard responsiveness for finance, claims, and operational reporting users.

Stack: SQL Server 2016/2017, Power BI, Power Query, DAX, GitHub Actions, SSMS, JIRA, Confluence, Databricks, Airflow DAGs

Database Developer | *Addison, TX (Client/s; Bank of America)*

Nov 2019 – Apr 2021

- Built and deployed SSIS packages for multi-source data extraction and transformation, ingesting from Excel, flat files, and relational databases with Conditional Split, Fuzzy Lookup, and Multicast for data scrubbing and staging validation.
- Designed automated ETL processes ensuring end-to-end data consistency, quality, and scalability across multiple operational systems.
- Investigated and resolved performance bottlenecks in stored procedures and complex SQL queries, improving execution plans and reducing load times across key reporting pipelines.

Stack: SQL Server 2016/2017, SSIS, SSRS, SSAS, Power BI, Visual Studio 2017, JIRA, Confluence, Bitbucket

Database Developer | *Highlands Ranch, CO (Client/s; Visa Inc, Evolent Health, Vintech)*

Aug 2016 – Oct 2019

- Maintained referential integrity and data consistency using SQL constraints; reviewed and refactored complex multi-join queries to improve performance across high-transaction financial datasets.
- Designed automated ETL processes ensuring end-to-end data consistency, quality, and scalability across multiple operational systems.

Stack: SQL Server 2016, SSIS, SSRS, Visual Studio 2017, JIRA, Confluence, Bitbucket

SKILLS

- **Cloud, Platforms & Engineering:** Azure Data Factory, Databricks, Synapse Analytics, ADLS Gen2, Blob Storage, Logic Apps, DevOps, Microsoft Fabric, OneLake, Snowflake, Power BI, ETL/ELT Pipelines, Delta Lake, Medallion Architecture, Data Lakehouse, Data Warehousing, SSIS Modernization, Spark Structured Streaming, Lakehouse Federation, Airflow, DAG Design, SLA Monitoring, Backfills, Dependency Management, dbt.
- **Programming & Optimization:** Python, PySpark, SQL, T-SQL, DAX, Spark SQL, Z-Ordering, Compaction, Caching, OPTIMIZE/MERGE, Data Skipping, Index Tuning, Query Tuning, Cost Optimization.
- **AI/ML, DevOps & Governance:** Generative AI, LLM Data Pipelines, MLflow, Feature Engineering, AI Agent Orchestration, Git, GitHub, Azure DevOps CI/CD, GitHub Actions, Databricks Notebooks, JIRA, Confluence, ServiceNow, HIPAA/PHI Masking, Access Controls, Schema Enforcement, Audit Logging, SCDs, Data Quality Validation.

EDUCATION

M.S. Information Technology | GPA: 3.8/4.00 | Campbellsville University, Kentucky

M.S. Computer Science | GPA: 3.7/4.00 | New York Institute of Technology, Manhattan, NY

B.S. Information Technology | GPA: 3.5/4.00 | Scient Institute of Technology, India